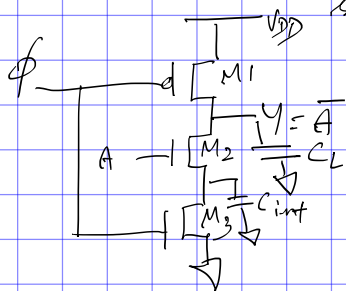


CMOS & pseudo NMOS: static gates

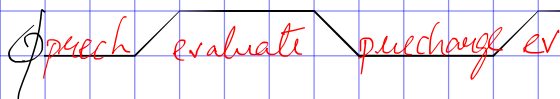
outputs are held at respective logic levels as long as they are powered.

Dynamic gates: Gates that utilize the charge stored in intermediate caps to retain logic levels. There is no path to V_{DD} or GND during operation.

Consider the dynamic inverter connected to a switching input ϕ .



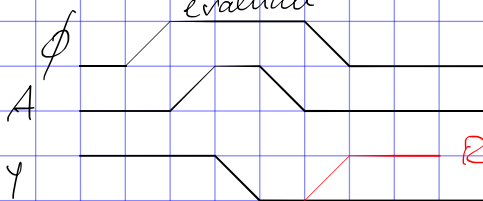
Phases of ϕ :



During the precharge phase of ϕ , M_1 is ON & M_3 is off. Output capacitance C_L is charged to V_{DD} . If $A=0$, there is charge sharing and the intermediate cap C_{int} which is charged to V_{DD} as well.

During the evaluate phase, $A=1 \Rightarrow C_L$ discharges to 0. $A=0 \Rightarrow C_L$ remains at V_{DD} .

Issue with multiple transitions at input.



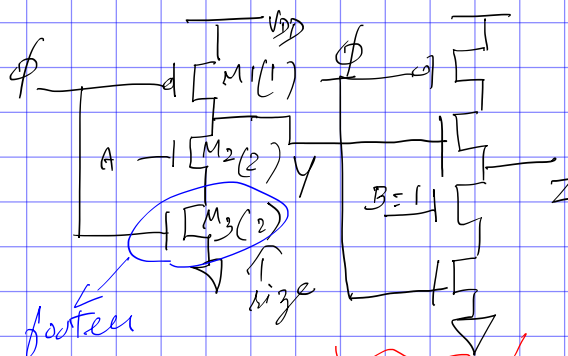
Required behaviour
Actual behaviour

Once C_L has discharged during the evaluate phase, the pull up M1 is disconnected and hence cannot be charged again. This is captured in the following table.

A
low
high
L $\rightarrow$ H
H $\rightarrow$ L

Y
high
low
H $\rightarrow$ L
not allowed.

} this is the monotonicity property of dynamic circuit.



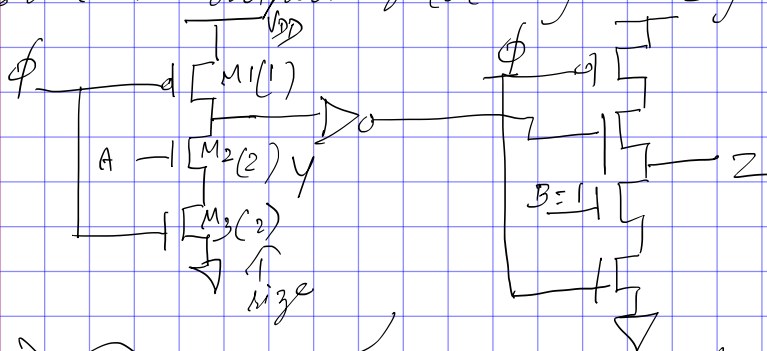
Absence of the large PUN makes the dynamic gates compact and fast but has a disadvantage of clock routing to all gates.

Dynamic NAND

When two dynamic gates are connected the monotonicity property leads to a contention.

The node Y can either be high or low or transition H $\rightarrow$ L. The succeeding dynamic NAND prohibits H $\rightarrow$ L transition at input.

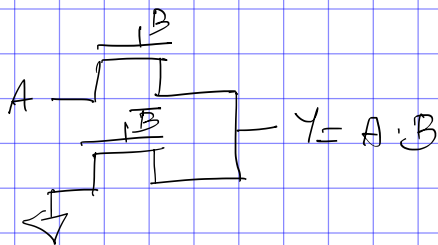
To overcome this drawback a CMOS inverter is used at the output of each dynamic gate.



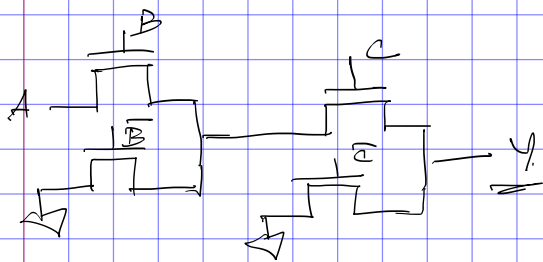
Domino logic - transition domino every cycle
The inverter at output implies gates are by default non-inverting.

Extending the usage of intermediate caps to store logic values:

Pass transistor logic



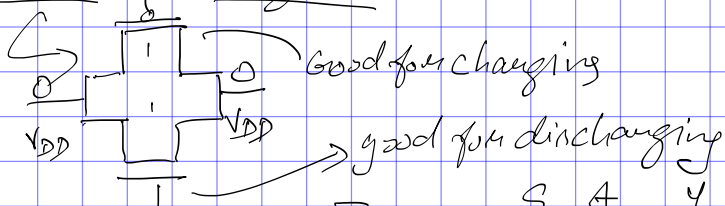
A	B	Y
V_{DD}	V_{DD}	$V_{DD} - V_{Tn}$
V_{DD}	0	0
0	V_{DD}	0
0	0	0



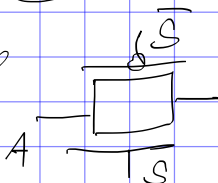
A	B	C	Y
V_{DD}	V_{DD}	V_{DD}	$V_{DD} - 2V_{Tn}$

Chaining pass transistors is infeasible beyond a couple of stages since the output voltage keeps dropping. CMOS gates can be used in between to restore logic levels.

To overcome the voltage drop issue, transmission gates are used.



Symbol

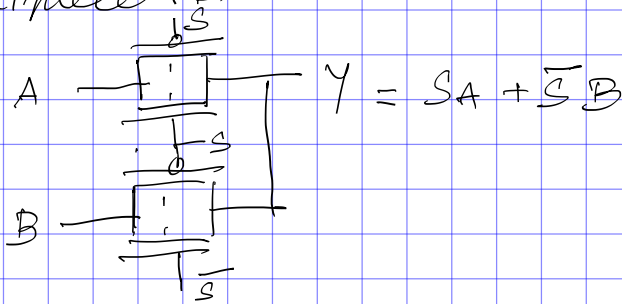


S	A	Y
V_{DD}	0	0
V_{DD}	V_{DD}	V_{DD}
0	0	0
0	V_{DD}	V_{DD}

$Z \rightarrow$ High impedance

Transmission gates are quite commonly used to construct multiplexers.

Multiplexers

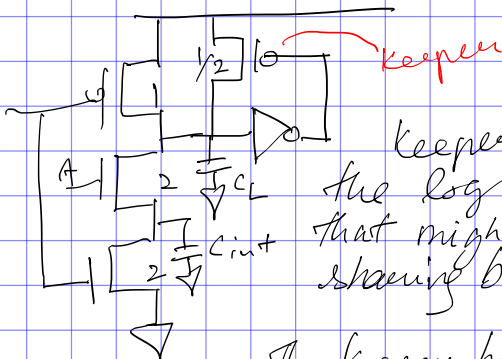


During charging both PMOS and NMOS are ON until V_{out} reaches $V_{DD} - V_{tn}$ after which PMOS alone takes over.

Similar phenomenon during discharging.

Since both devices are not on for the entirety we will assume that the R_{eq} of a transmission gate $\approx R_{eqn}$.

Keepers



Keeper is used to restore the logic level of the output that might drop due to charge sharing between C_L & C_{int} .

The keeper has to be weak compared to the PDN so that it allows discharging C_L when necessary.

HW populate the logical effort table:

	inverter		n-input NAND		n-input NOR	
	P _{DN}	P _{UN}	P _{DN}	P _{UN}	P _{DN}	P _{UN}
Static CMOS						
H _l skew						
L _o skew						
Pseudo nMOS						
Dynamic logic						